MINEX III Report Card

Template Generator cogent+0515



Last Updated: November 10, 2026

Participant Details

Company: Thales
Provided CBEFF PID: 0017 0515
Associated Submission Identifiers: gemalto

Date Application Received: 12/11/2023
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Library	Size (bytes)	MD5 Checksum
libonnxruntime.so.1.11.1	15086128	8edc77c21cebb20d414601ca63c4a9de
libCgtBAX_ORT_CPU.so	35168	7202596d5370cd1143ab97e973a58f29
libminexiii_cogent_0515.so	369408	c93f7eda977df232f808a91c446d8fe8
finger_v4.dat	59077852	a2ddf691cd4f8aa45bd1bb3b55e40f1f

Compliance Test Results

The following presents **PIV compliance** results per the criteria detailed in NIST Special Publication 800-76-2: Biometric Specifications for Personal Identity Verification.

It also includes **MINEX III compliance** results per the criteria detailed in sections 4 through 8 of the Minutia Interoperability Exchange (MINEX) III Test Plan and Application Programming Interface.

PIV: PASS
- All certified matchers must be able to match templates from this template generator with an $\text{FNMR}_{\text{FMR}}(0.01) \leq 0.01$ using two fingers (4.5.2.2-3). ✓
- Average template creation time must be no more than 500 milliseconds (4.5.2.2-2). ✓
- Minutia density plots derived from generated templates do not exhibit a periodic, grid-like, or geometric structure. ✓
- If matcher also submitted, matcher is PIV Level Two compliant. ✓

MINEX III: PASS
- Must pass MINEX III validation. ✓
- Must be PIV compliant. ✓
- No more than two compliant template generators from the submitting organization, or its subsidiaries, acquisitions, or mergers allowed (8.8). ✓
- If matcher also submitted, matcher is MINEX III compliant. ✓

Notes

- This report will be updated as new matching algorithms and template generators pass the compliance test. These updates will not change the PASS/FAIL decision above.
- NIST reserves the right to decertify a template generator if it later discovers the template generator violates MINEX III or PIV specifications in some previously undetected way.
- This is the "*best*" compliant submission from Thales, and is therefore a member of the pooled DET curves published throughout all MINEX III report cards.

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1 Introduction

Testing is performed at a NIST facility. Each participant's submission is validated by NIST before undergoing full testing to ensure it operates correctly. If the matcher passes the validation procedure, it is then used to compare standard fingerprint templates. Performance is assessed against templates created by a template generator submitted by the participant as well as templates created by other MINEX III compliant template generators.

2 Methodology

Testing is performed at a NIST facility. Each participant's submission is validated by NIST before undergoing full testing to ensure it operates correctly. If the template generator passes the validation procedure, performance is assessed by using MINEX III compliant matching algorithms to compare templates created by the template generator. These matchers were submitted to the ongoing MINEX III program by various participants.

2.1 Dataset

Testing is performed over a single dataset of sequestered fingerprint images. The images were collected by U.S. Visit at ports of entry into the United States. They consist of Live-scan plain impressions of left and right index fingers. WSQ [1] compression was applied to all images at a ratio of 15:1. The most recent capture of each subject was treated as the authentication sample, and the next most recent as the enrolled sample.

The dataset was divided into $533\,767$ mated and $1\,067\,530$ non-mated subject pairings. Since both left and right index fingerprints are available for each subject, this provides $1\,061\,657$ mated and $2\,127\,712$ non-mated single-finger comparisons (after database consolidation). When left and right index fingers are fused at the score level [3, 8], the sets condense to $530\,394$ mated and $1\,062\,814$ non-mated comparison scores.

2.2 Accuracy Metrics

Core matching accuracy is presented in the form of Detection Error Tradeoff (DET) plots [7], which show the trade-off between the False Match Rate (FMR) and the False Non-Match Rate (FNMR) as a decision threshold is adjusted. Formally, let m_i ($i = 1 \ldots M$) be the ith mated comparison score, and n_j ($j = 1 \ldots N$) the jth non-mated comparison score. Then the statistics are

$$\mathrm{FMR}(\tau) = \frac{1}{N} \sum_{j=1}^{N} \mathbb{1}\{n_j \geq \tau\}, \tag{1}$$

$$\mathrm{FNMR}(\tau) = \frac{1}{M} \sum_{i=1}^{M} \mathbb{1}\{m_i < \tau\}. \tag{2}$$

where $\mathbb{1}\{A\}$ is the indicator [4] of event A. Equations 1 and 2 define the curve parametrically with the decision threshold, τ, as the free parameter. In some figures and tables, FNMR is presented as a function of FMR. This relationship is determined by

$$\mathrm{FNMR}_{\mathrm{FMR}}(\alpha) = \min_{\tau} \{\ \mathrm{FNMR}(\tau) \mid \mathrm{FMR}(\tau) \leq \alpha\ \}, \tag{3}$$

which reads as the smallest FNMR that can be achieved while maintaining an FMR less than or equal to α, the targeted FMR. This method of relating the two error statistics ensures FNMR is well-defined for all $0 \leq \alpha \leq 1$. It also imposes a natural penalty on matching algorithms that produce heavily discretized scores.

2.3 Uncertainty Estimation

Some figures in this report include boxplots that convey the uncertainty associated with a statistic. The boxplots are intended to show the expected variation in the observed value if one assumes repeated iid sampling from the same population. They are not intended to reflect how the statistic might change over different test data or even different sampling strategies over the same data.

Estimates of uncertainty are computed using the Wilson Score method [10] which overcomes certain problems associated with applying the Central Limit Theorem to a discretized estimator. We make several simplifying assumptions when applying the method to biometric identification. Most notably, separate searches against the same enrollment database are treated as independent samples, yet we know positive correlations exist due to Doddingtons Zoo [2]. We also report estimates of the variability of FNIR at a fixed FPIR when in fact it is the decision threshold that is fixed. Uncertainty with respect to what decision threshold corresponds to the targeted FPIR results in increased uncertainty about the true value of FNIR. However, our estimates of FPIR are fairly

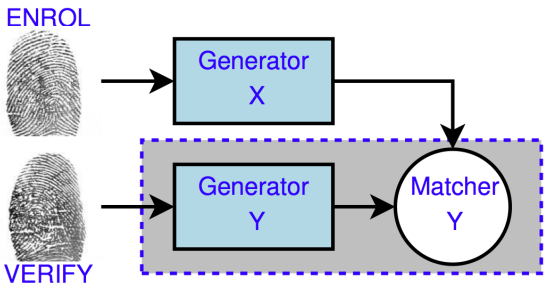


Figure 1: MINEX III Interoperability Test Setup

tight due to the large number of non-mated searches performed, so they are not expected to have a large impact on the estimates.

2.4 Interoperability

Interoperability is tested in a manner similar to *Scenario 1* from the MINEX Evaluation Report [5] (see Figure 1). An enrolment template is prepared using submission X. Submission Y is used to prepare the authentication template and perform the match. The authentication template is always prepared by the same submission used to compare the templates. However, enrolment templates need not originate from the same submission. When they do, we refer to as ”native” mode.

3 Results

This section details the performance of template generator cogent+0515. Sections 3.1 and 3.2 present accuracy results for single finger and two finger matching respectively. Section 3.4 presents information on the number of minutia the template generator finds in the samples.

3.1 Single Finger

Singe finger comparison results show the combined results for left and right index comparisons. For reference, *NIST Special Publication 800-76-2* requires that the template generator achieve an accuracy of $\text{FNMR}_{\text{FMR}}(0.01) \leq 0.01$ against all compliant matchers.

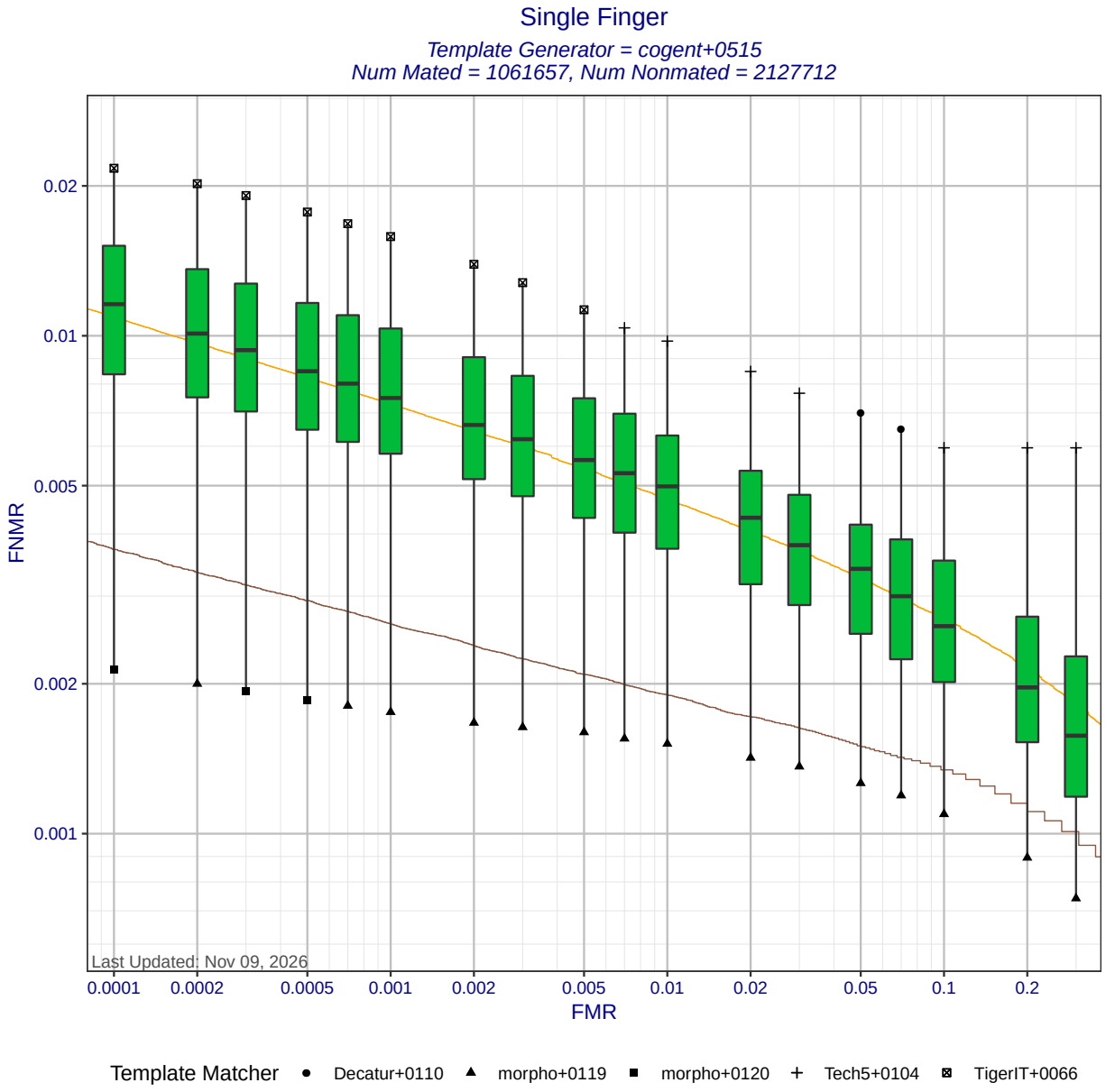


Figure 2: *Single finger DET statistics for template generator cogent+0515. Each box shows the distribution of FNMRs at a fixed FMR across different matchers. The whisker ends show the minimum and maximum FNMRs. The brown curve shows the DET curve when the matcher and template generators were submitted by the same participant. The orange DET curve shows pooled performance when all matchers compare templates created by cogent+0515.*

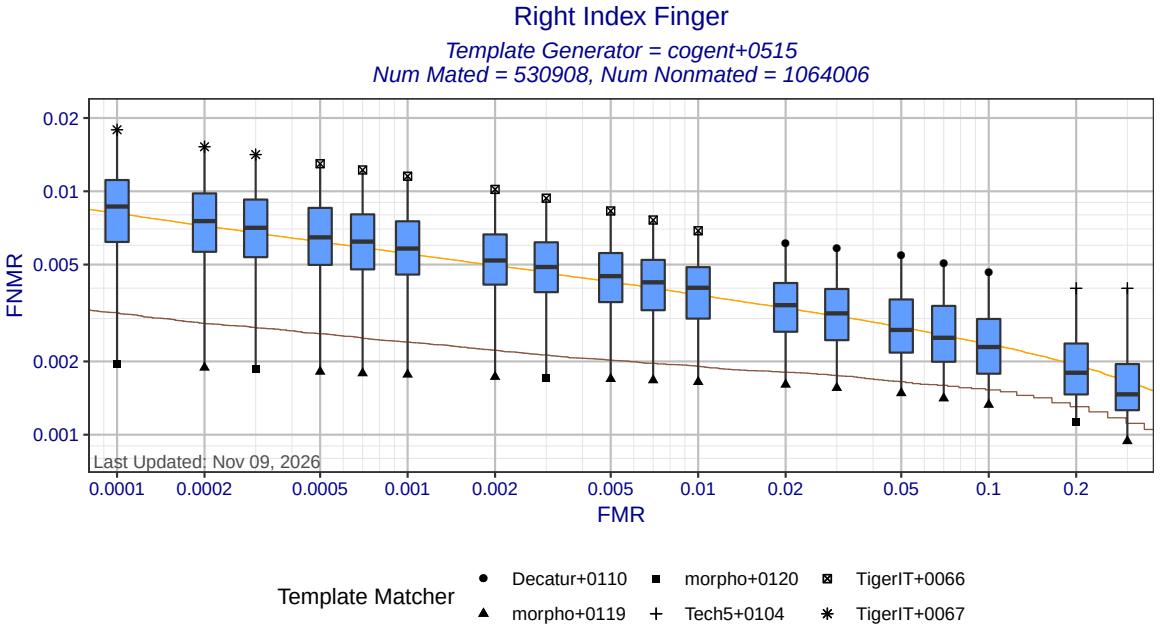


Figure 3: *Right Index Finger DET statistics for template generator cogent+0515. Each box shows the distribution of FNMRs at a fixed FMR across different matchers. The whisker ends show the minimum and maximum FNMRs. The brown curve shows the DET curve when the matcher and template generators were submitted by the same participant. The orange DET curve shows pooled performance when all matchers use templates created by cogent+0515.*

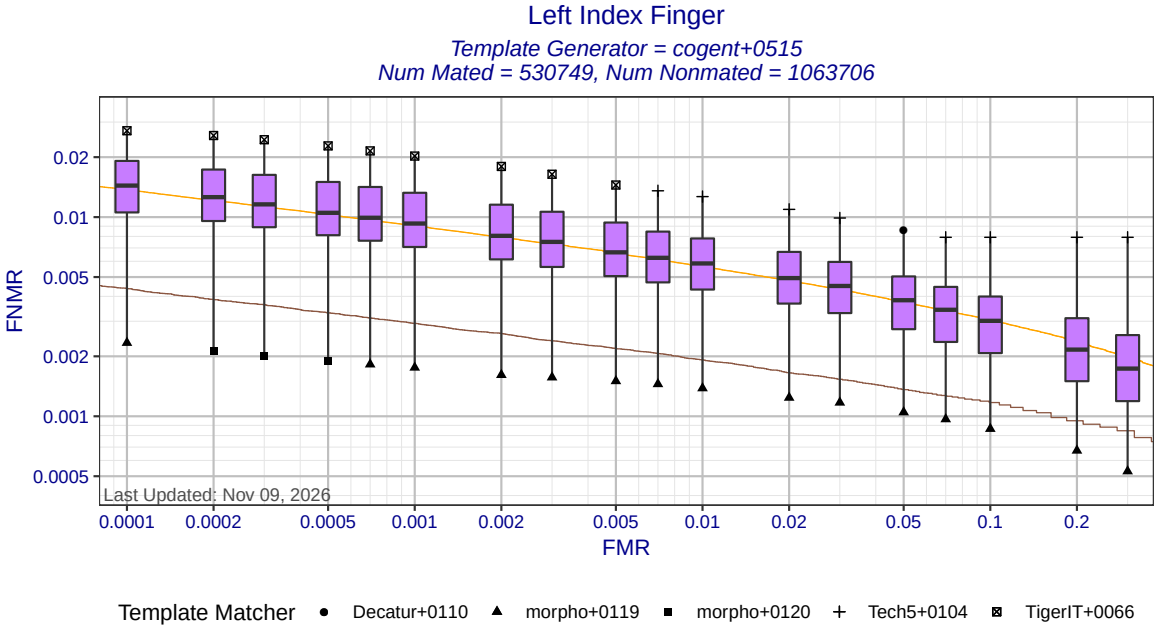


Figure 4: *Left Index Finger DET statistics for template generator cogent+0515. Each box shows the distribution of FNMRs at a fixed FMR across different template generators. The brown curve shows the DET curve when the matcher and template generators were submitted by the same participant. The orange DET curve shows pooled performance when all matchers use templates created by cogent+0515.*

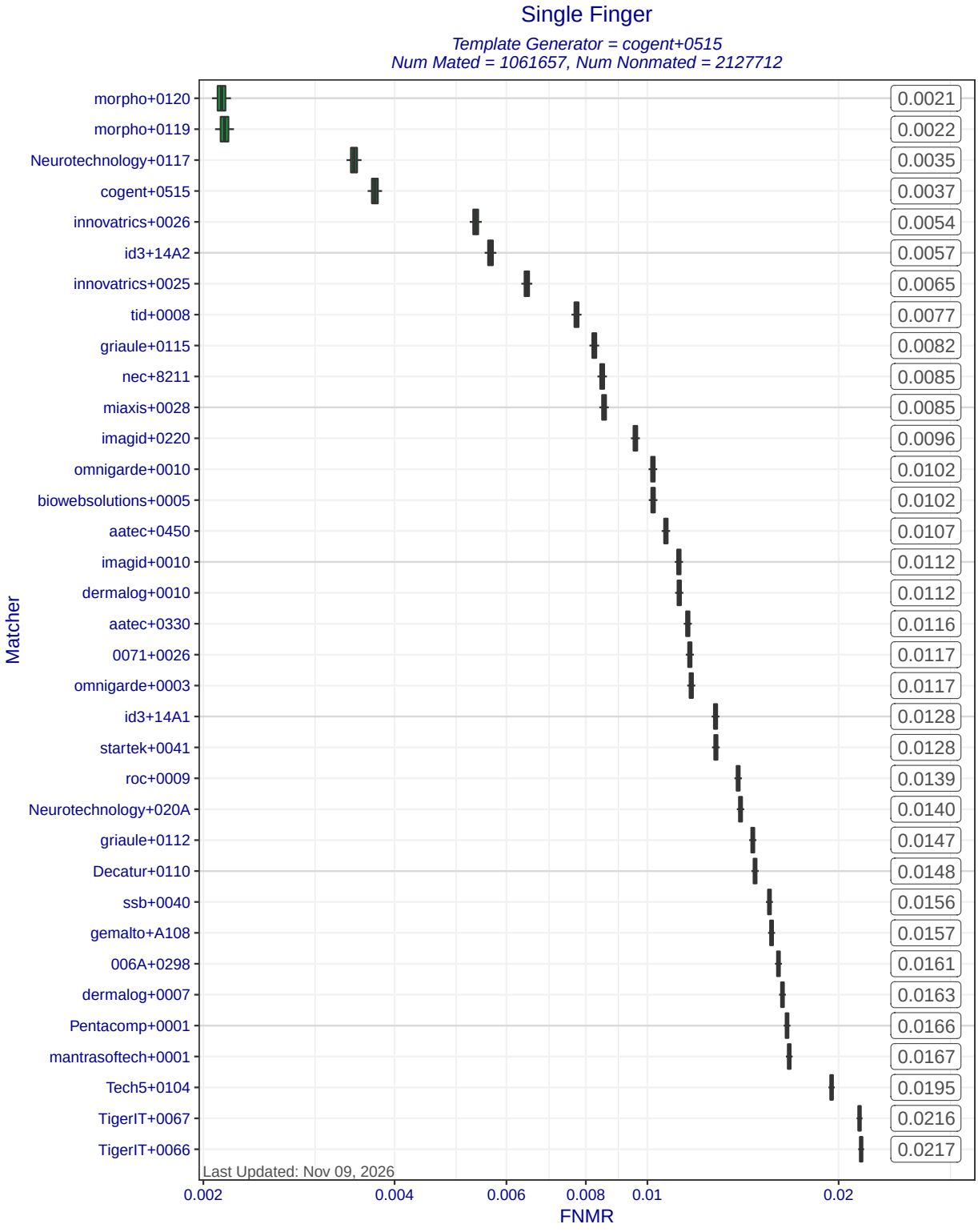


Figure 5: *Single Finger FNMRs at FMR=0.0001 when MINEX III compliant matchers compare templates created by template generator cogent+0515. Each box represents uncertainty about the true FNMR. The box edges mark the 50% confidence intervals while the whiskers mark the 90% confidence intervals. The numbers on the right show the actual computed FNMRs.*

3.2 Two Finger

This section presents accuracy when different MINEX III compliant matchers compare templates created by template generator cogent+0515. Two finger fusion is achieved by averaging the scores for left and right index fingers for each person.

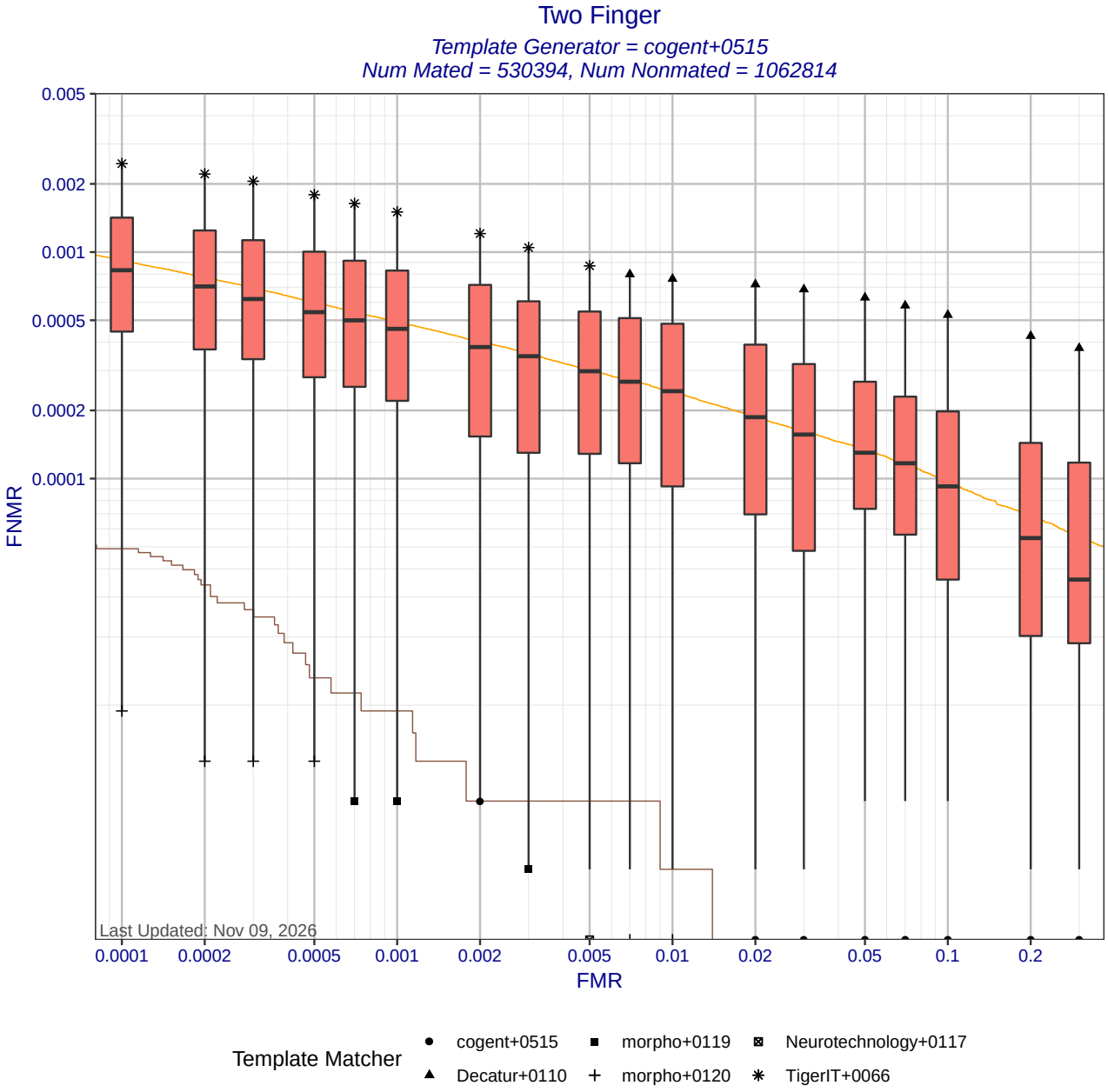


Figure 6: *Two Finger DET statistics for template generator cogent+0515. Each box shows the distribution of FNMRs at a fixed FMR across different matchers. The whisker ends show the minimum and maximum FNMRs. The brown curve shows the DET curve when the matcher and template generators were submitted by the same participant. The orange DET curve shows pooled performance when all matchers use templates created by cogent+0515. Score-level fusion is achieved by averaging the scores for left and right index fingers.*

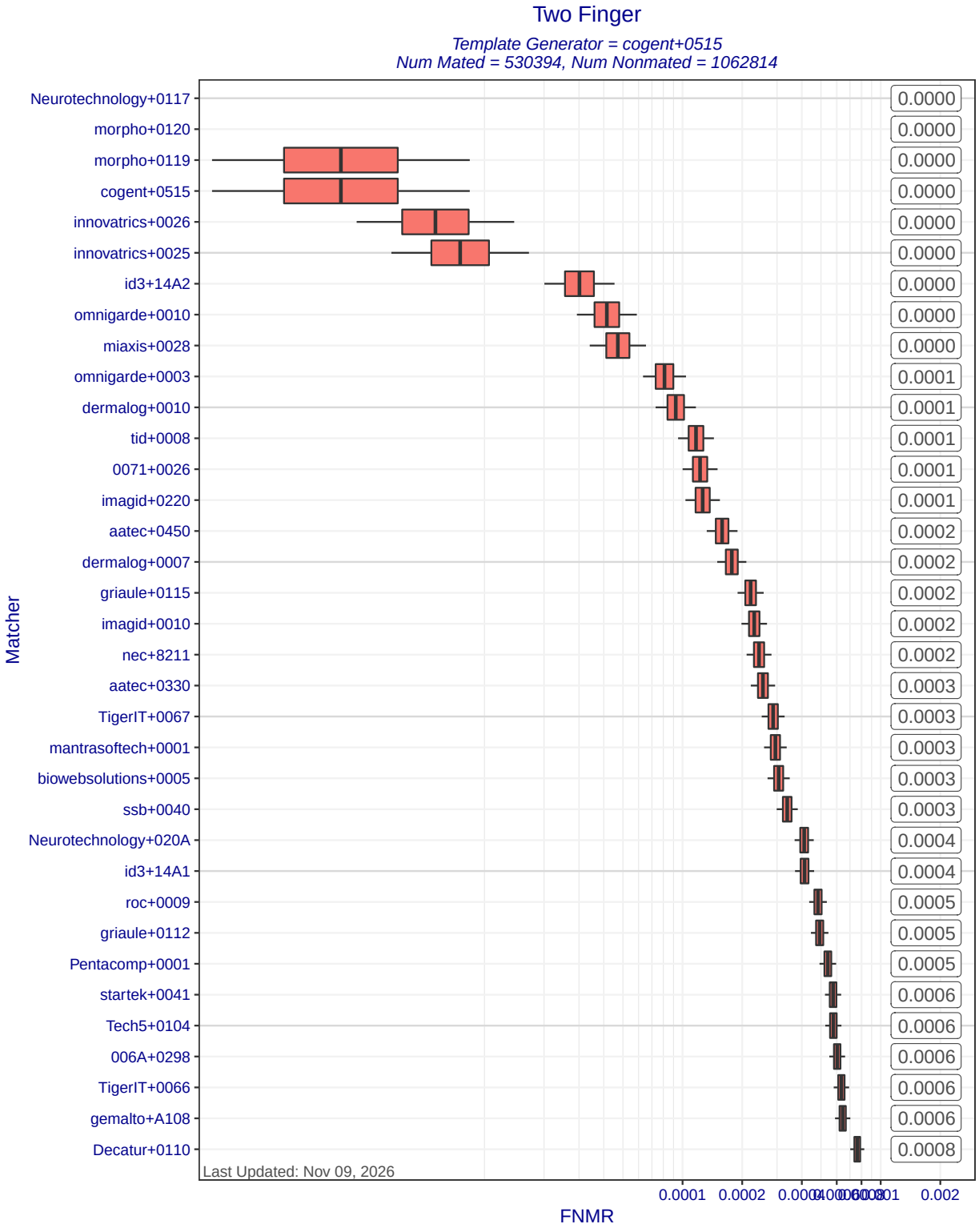


Figure 7: *Two Finger FNMR at FMR=0.01 when different matchers compare templates created by template generator cogent+0515. Each box represents uncertainty about the true FNMR. The box edges mark the 50% confidence intervals while the whiskers mark the 90% confidence intervals. The numbers on the right show the actual computed FNMRs. Score-level fusion is achieved by averaging the scores for left and right index fingers.*

3.3 Template Creation Times

To achieve PIV compliance, the template generator must create templates in no more than 0.5 seconds (500 milliseconds) on average.

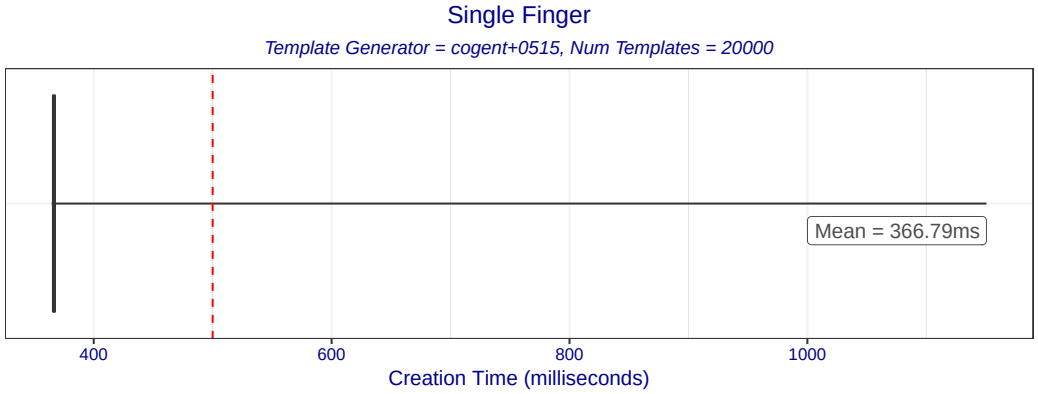


Figure 8: *Boxplot of template creation times for template generator cogent+0515. The box edges mark the 10th and 90th percentiles while the whiskers mark the maximum and minimum creation times.*

3.4 Minutia Counts

This section presents information relating to the number of minutia the template generator finds in fingerprint images. The relative number of minutia found in common fingerprint images has been shown to influence matching outcomes [9, 6].

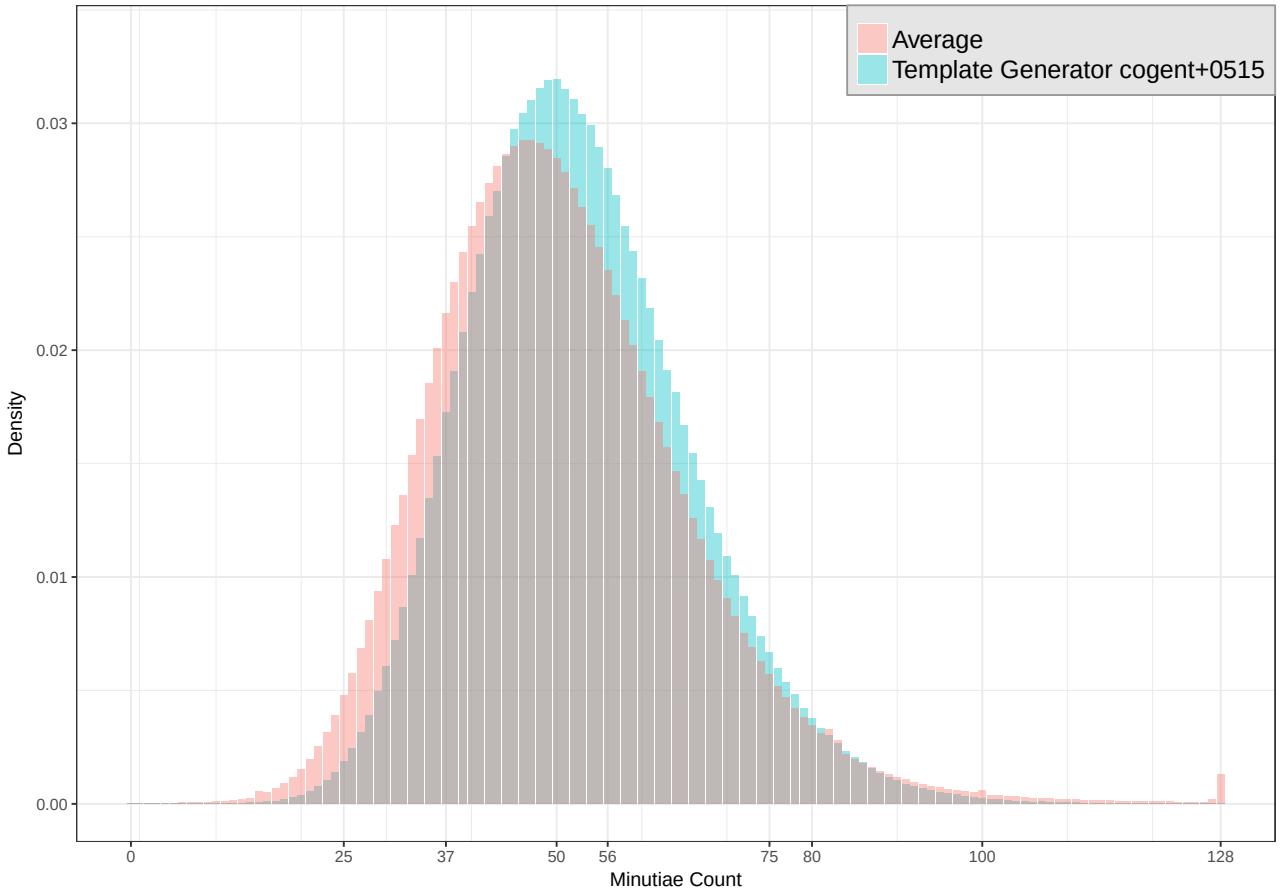


Figure 9: *Probability distribution of the number of minutia the template generator found in the samples. The* average *probability distribution shows the combined distribution of minutia counts across all compliant template generators submitted for MINEX III (i.e., excluding Ongoing MINEX template generators).*

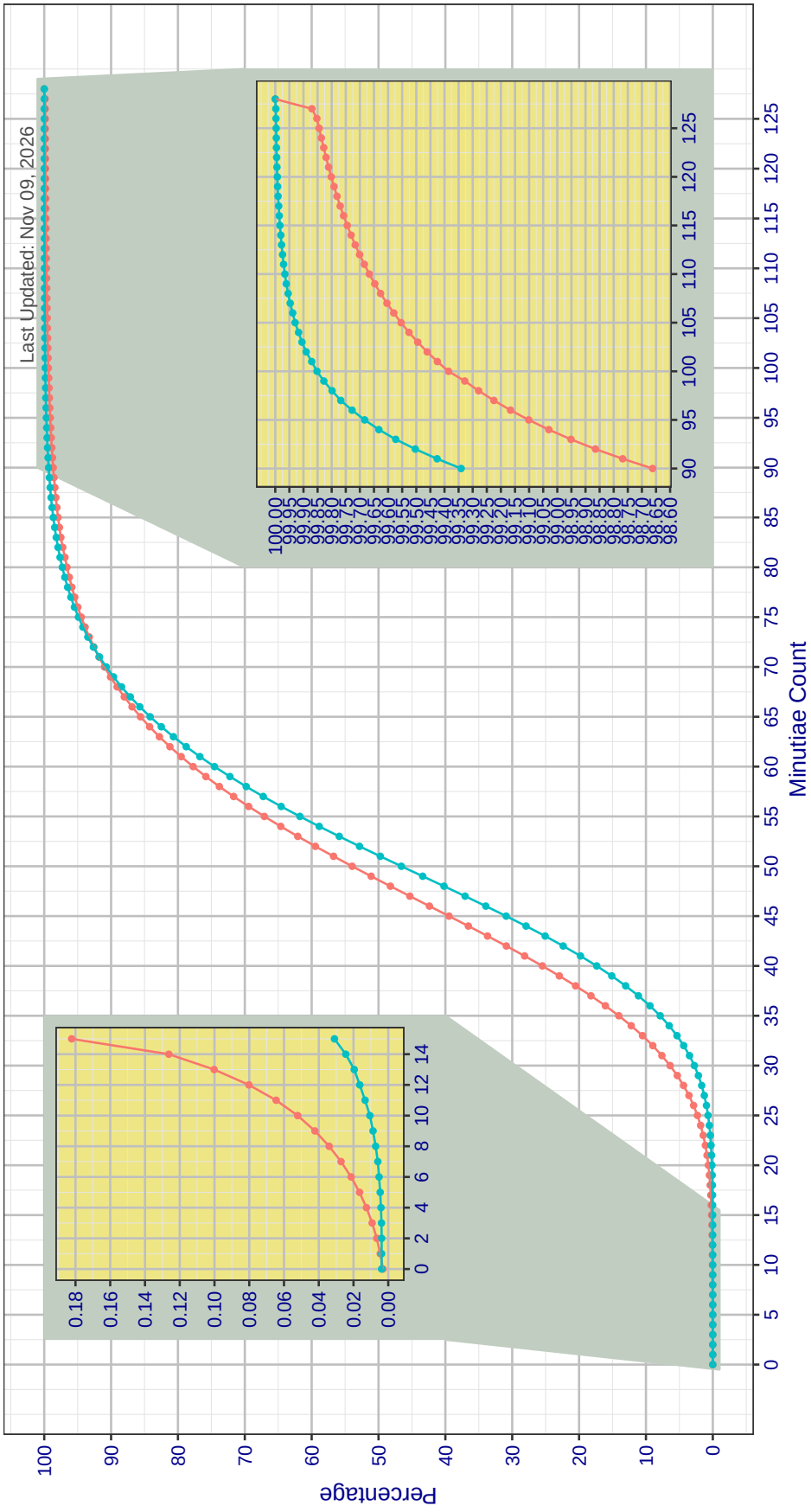


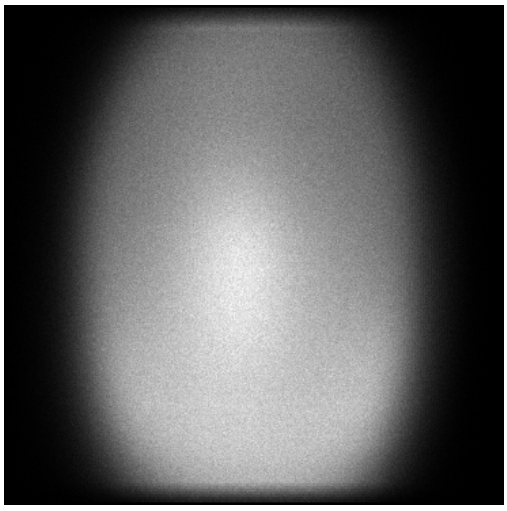
Figure 10: *Cumulative summation of the number of minutia the template generator found in the samples. The* average *probability distribution shows the combined distribution of minutia counts across all compliant template generators submitted for MINEX III (i.e., excluding Ongoing MINEX template generators).*

3.5 Minutia Density Plots

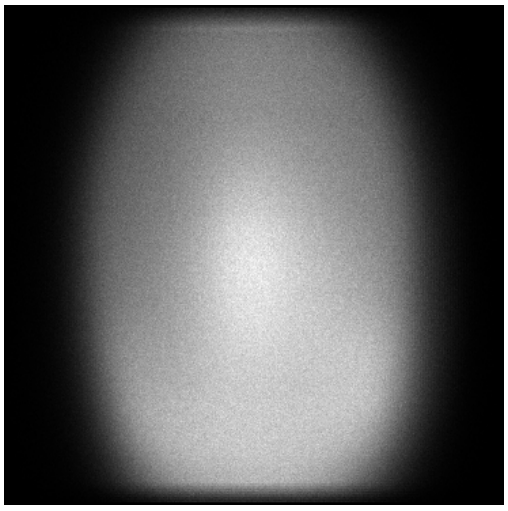
Minutia density plots show where the template generator tends to find minutia in fingerprint images. They are 2D histograms where the degree of illumination at an (x, y) coordinate indicates how frequently the software located a minutiae point at that location. The purpose of showing minutia density plots is to determine whether the template generator exhibits regional preference when locating minutia.

Some template generators produce minutia that exhibit a periodic structure, but this template generator does **not**. Periodic structures and other regional preferences are an indication that the template generator is departing from the minutia placement requirements of INCITS 378, clause 5. The expected pattern is a locally uniform distribution, and the appearance of local structure indicates systematic non-conformance with the standard. Given such behavior negatively affects interoperability[9], developers are asked to determine the cause of such behavior – for example, as an artifact of a tilebased image processing algorithms applied to the input fingerprint image – and to resubmit corrected algorithms.

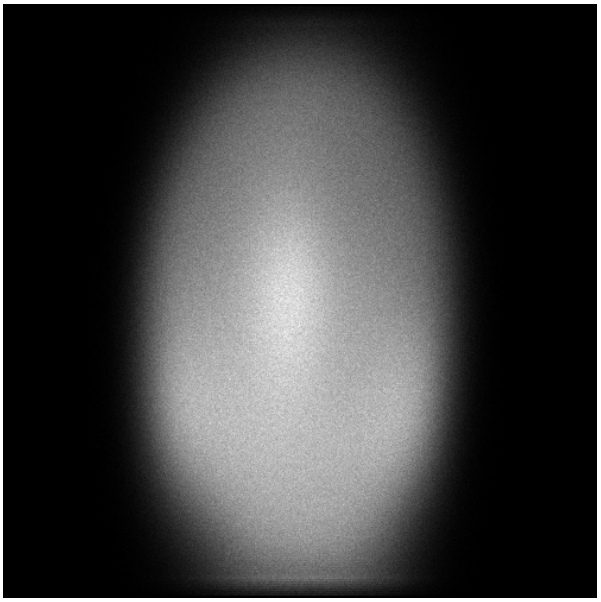
NIST uses a closed-form test to detect high frequency periodic structure by searching for modulation in the 368x368 minutia plot's Fourier reprentation. The code for this test is available on GitHub.



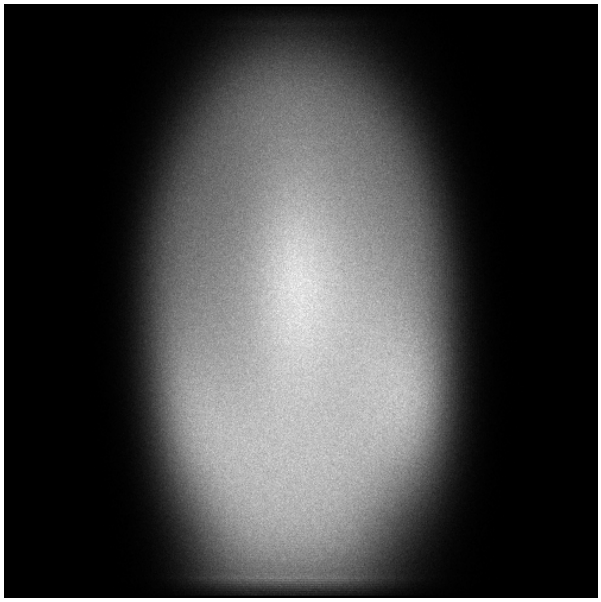
(a) Minutia density plot for 671 619 368x368 left indexes.



(b) Minutia density plot for 671 460 368x368 right indexes.



(c) Minutia density plot for 477 312 500x500 left indexes.



(d) Minutia density plot for 477 317 500x500 right indexes.

Figure 11: 2D Minutia Placement Density Functions.

3.6 Comparison to Ongoing MINEX

MINEX III uses a larger set of comparisons than the older ongoing MINEX evaluation. Although this is generally good because it provides more accurate estimates of performance in MINEX III, it makes it more difficult to directly compare the results in this report to the archived ones from ongoing MINEX. The tables below report DET accuracy at fixed FMRs computed over the same set of comparisons that were used in ongoing MINEX. Ongoing MINEX reported FNMR at FMR $= 0.01$ for two-finger.

Table 1: *Single finger FNMRs at various FMRs when cogent+0515 and MINEX III-compliant matchers compare templates created by template generator cogent+0515.*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
006A+0298	0.0066 ± 0.0003	0.0109 ± 0.0003	0.0159 ± 0.0004
0071+0026	0.0038 ± 0.0002	0.0069 ± 0.0003	0.0106 ± 0.0003
aatec+0330	0.0042 ± 0.0002	0.0070 ± 0.0003	0.0111 ± 0.0003
aatec+0450	0.0037 ± 0.0002	0.0066 ± 0.0003	0.0108 ± 0.0003
biowebsolutions+0005	0.0046 ± 0.0002	0.0074 ± 0.0003	0.0106 ± 0.0003
cogent+0515	0.0012 ± 0.0001	0.0020 ± 0.0001	0.0033 ± 0.0002
Decatur+0110	0.0089 ± 0.0003	0.0119 ± 0.0004	0.0173 ± 0.0004
dermalog+0007	0.0071 ± 0.0003	0.0122 ± 0.0004	0.0193 ± 0.0005
dermalog+0010	0.0060 ± 0.0003	0.0100 ± 0.0003	0.0148 ± 0.0004
gemalto+A108	0.0070 ± 0.0003	0.0108 ± 0.0003	0.0155 ± 0.0004
griaule+0112	0.0064 ± 0.0003	0.0103 ± 0.0003	0.0157 ± 0.0004
griaule+0115	0.0040 ± 0.0002	0.0057 ± 0.0002	0.0084 ± 0.0003
id3+14A1	0.0051 ± 0.0002	0.0082 ± 0.0003	0.0134 ± 0.0004
id3+14A2	0.0022 ± 0.0002	0.0036 ± 0.0002	0.0056 ± 0.0002
imagid+0010	0.0045 ± 0.0002	0.0076 ± 0.0003	0.0123 ± 0.0004
imagid+0220	0.0034 ± 0.0002	0.0061 ± 0.0003	0.0093 ± 0.0003
innovatrics+0025	0.0021 ± 0.0002	0.0039 ± 0.0002	0.0070 ± 0.0003
innovatrics+0026	0.0020 ± 0.0001	0.0035 ± 0.0002	0.0058 ± 0.0003
mantrasoftech+0001	0.0052 ± 0.0002	0.0096 ± 0.0003	0.0164 ± 0.0004
miaxis+0028	0.0028 ± 0.0002	0.0051 ± 0.0002	0.0081 ± 0.0003
morpho+0119	0.00090 ± 0.00010	0.0013 ± 0.0001	0.0019 ± 0.0001
morpho+0120	0.0009 ± 0.0001	0.0013 ± 0.0001	0.0018 ± 0.0001
nec+8211	0.0033 ± 0.0002	0.0052 ± 0.0002	0.0078 ± 0.0003
Neurotechnology+0117	0.0011 ± 0.0001	0.0019 ± 0.0001	0.0032 ± 0.0002
Neurotechnology+020A	0.0051 ± 0.0002	0.0090 ± 0.0003	0.0145 ± 0.0004
omnigarde+0003	0.0053 ± 0.0002	0.0086 ± 0.0003	0.0132 ± 0.0004
omnigarde+0010	0.0045 ± 0.0002	0.0076 ± 0.0003	0.0119 ± 0.0004
Pentacomp+0001	0.0059 ± 0.0003	0.0106 ± 0.0003	0.0178 ± 0.0004
roc+0009	0.0051 ± 0.0002	0.0085 ± 0.0003	0.0154 ± 0.0004
ssb+0040	0.0066 ± 0.0003	0.0113 ± 0.0003	0.0195 ± 0.0005
startek+0041	0.0056 ± 0.0002	0.0087 ± 0.0003	0.0137 ± 0.0004
Tech5+0104	0.0111 ± 0.0003	0.0160 ± 0.0004	0.0228 ± 0.0005
tid+0008	0.0030 ± 0.0002	0.0051 ± 0.0002	0.0076 ± 0.0003
TigerIT+0066	0.0126 ± 0.0004	0.0209 ± 0.0005	0.0269 ± 0.0005
TigerIT+0067	0.0086 ± 0.0003	0.0177 ± 0.0004	0.0284 ± 0.0005

Table 2: *Two finger FNMRs at various FMRs when cogent+0515 and MINEX III-compliant matchers compare templates created by template generator cogent+0515.*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
006A+0298	0.0005 ± 0.0001	0.0010 ± 0.0001	0.0016 ± 0.0002
0071+0026	0.00012 ± 0.00005	0.00034 ± 0.00009	0.0005 ± 0.0001
aatec+0330	0.00019 ± 0.00007	0.0005 ± 0.0001	0.0008 ± 0.0001

Table 2: *(continued)*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
aatec+0450	0.00020 ± 0.00007	0.00038 ± 0.00009	0.0007 ± 0.0001
biowebsolutions+0005	0.00034 ± 0.00009	0.0006 ± 0.0001	0.0008 ± 0.0001
cogent+0515	0.00001 ± 0.00002	0.00002 ± 0.00002	0.00007 ± 0.00004
Decatur+0110	0.0008 ± 0.0001	0.0011 ± 0.0002	0.0018 ± 0.0002
dermalog+0007	0.00015 ± 0.00006	0.0005 ± 0.0001	0.0012 ± 0.0002
dermalog+0010	0.00026 ± 0.00008	0.0007 ± 0.0001	0.0012 ± 0.0002
gemalto+A108	0.0006 ± 0.0001	0.0010 ± 0.0001	0.0014 ± 0.0002
griaule+0112	0.0005 ± 0.0001	0.0008 ± 0.0001	0.0014 ± 0.0002
griaule+0115	0.00024 ± 0.00007	0.00043 ± 0.00010	0.0007 ± 0.0001
id3+14A1	0.00037 ± 0.00009	0.0007 ± 0.0001	0.0010 ± 0.0001
id3+14A2	0.00007 ± 0.00004	0.00014 ± 0.00006	0.00026 ± 0.00008
imagid+0010	0.00023 ± 0.00007	0.0005 ± 0.0001	0.0009 ± 0.0001
imagid+0220	0.00016 ± 0.00006	0.00040 ± 0.00009	0.0006 ± 0.0001
innovatrics+0025	0.00004 ± 0.00003	0.00009 ± 0.00005	0.00031 ± 0.00008
innovatrics+0026	0.00003 ± 0.00003	0.00009 ± 0.00005	0.00024 ± 0.00007
mantrasoftech+0001	0.00023 ± 0.00007	0.0007 ± 0.0001	0.0018 ± 0.0002
miaxis+0028	0.00008 ± 0.00004	0.00025 ± 0.00007	0.00042 ± 0.00010
morpho+0119	0.00002 ± 0.00003	0.00002 ± 0.00003	0.00003 ± 0.00003
morpho+0120	0.00002 ± 0.00003	0.00002 ± 0.00003	0.00003 ± 0.00003
nec+8211	0.00026 ± 0.00008	0.00041 ± 0.00010	0.0006 ± 0.0001
Neurotechnology+0117	0.00002 ± 0.00002	0.00003 ± 0.00003	0.00011 ± 0.00005
Neurotechnology+020A	0.00036 ± 0.00009	0.0006 ± 0.0001	0.0011 ± 0.0002
omnigarde+0003	0.00012 ± 0.00005	0.00032 ± 0.00008	0.0006 ± 0.0001
omnigarde+0010	0.00008 ± 0.00004	0.00030 ± 0.00008	0.0006 ± 0.0001
Pentacomp+0001	0.0005 ± 0.0001	0.0009 ± 0.0001	0.0020 ± 0.0002
roc+0009	0.0005 ± 0.0001	0.0008 ± 0.0001	0.0013 ± 0.0002
ssb+0040	0.00045 ± 0.00010	0.0009 ± 0.0001	0.0017 ± 0.0002
startek+0041	0.00045 ± 0.00010	0.0009 ± 0.0001	0.0015 ± 0.0002
Tech5+0104	0.0008 ± 0.0001	0.0012 ± 0.0002	0.0018 ± 0.0002
tid+0008	0.00016 ± 0.00006	0.00040 ± 0.00009	0.0007 ± 0.0001
TigerIT+0066	0.0012 ± 0.0002	0.0024 ± 0.0002	0.0034 ± 0.0003
TigerIT+0067	0.0007 ± 0.0001	0.0020 ± 0.0002	0.0045 ± 0.0003

4 Performance Tables

The following tables present accuracy numbers, including estimates of uncertainty in the form of 90% confidence bounds. These tables are provided because most of the figures in the main body of the report do not present actual accuracy numbers.

Table 3: *Single finger FNMRs at various FMRs when cogent+0515 and MINEX III-compliant matchers compare templates created by template generator cogent+0515.*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
006A+0298	0.0071 ± 0.0001	0.0108 ± 0.0002	0.0161 ± 0.0002
0071+0026	0.0043 ± 0.0001	0.0073 ± 0.0001	0.0117 ± 0.0002
aatec+0330	0.0050 ± 0.0001	0.0076 ± 0.0001	0.0116 ± 0.0002
aatec+0450	0.0043 ± 0.0001	0.0068 ± 0.0001	0.0107 ± 0.0002
biowebsolutions+0005	0.0053 ± 0.0001	0.0073 ± 0.0001	0.0102 ± 0.0002
cogent+0515	0.00190 ± 0.00007	0.00264 ± 0.00008	0.00372 ± 0.00010
Decatur+0110	0.0086 ± 0.0001	0.0109 ± 0.0002	0.0148 ± 0.0002
dermalog+0007	0.0065 ± 0.0001	0.0106 ± 0.0002	0.0163 ± 0.0002
dermalog+0010	0.0044 ± 0.0001	0.0074 ± 0.0001	0.0112 ± 0.0002
gemalto+A108	0.0075 ± 0.0001	0.0110 ± 0.0002	0.0157 ± 0.0002
griaule+0112	0.0066 ± 0.0001	0.0101 ± 0.0002	0.0147 ± 0.0002
griaule+0115	0.0045 ± 0.0001	0.0059 ± 0.0001	0.0082 ± 0.0001
id3+14A1	0.0057 ± 0.0001	0.0085 ± 0.0001	0.0128 ± 0.0002
id3+14A2	0.00277 ± 0.00008	0.0040 ± 0.0001	0.0057 ± 0.0001
imagid+0010	0.0050 ± 0.0001	0.0075 ± 0.0001	0.0112 ± 0.0002
imagid+0220	0.0040 ± 0.0001	0.0061 ± 0.0001	0.0096 ± 0.0002
innovatrics+0025	0.00246 ± 0.00008	0.0040 ± 0.0001	0.0065 ± 0.0001
innovatrics+0026	0.00229 ± 0.00008	0.00344 ± 0.00009	0.0054 ± 0.0001
mantrasoftech+0001	0.0057 ± 0.0001	0.0099 ± 0.0002	0.0167 ± 0.0002
miaxis+0028	0.00336 ± 0.00009	0.0055 ± 0.0001	0.0085 ± 0.0001
morpho+0119	0.00152 ± 0.00006	0.00175 ± 0.00007	0.00216 ± 0.00007
morpho+0120	0.00154 ± 0.00006	0.00177 ± 0.00007	0.00214 ± 0.00007
nec+8211	0.0043 ± 0.0001	0.0062 ± 0.0001	0.0085 ± 0.0001
Neurotechnology+0117	0.00164 ± 0.00006	0.00235 ± 0.00008	0.00345 ± 0.00009
Neurotechnology+020A	0.0058 ± 0.0001	0.0092 ± 0.0002	0.0140 ± 0.0002
omnigarde+0003	0.0046 ± 0.0001	0.0076 ± 0.0001	0.0117 ± 0.0002
omnigarde+0010	0.0041 ± 0.0001	0.0066 ± 0.0001	0.0102 ± 0.0002
Pentacomp+0001	0.0063 ± 0.0001	0.0106 ± 0.0002	0.0166 ± 0.0002
roc+0009	0.0057 ± 0.0001	0.0087 ± 0.0001	0.0139 ± 0.0002
ssb+0040	0.0066 ± 0.0001	0.0106 ± 0.0002	0.0156 ± 0.0002
startek+0041	0.0063 ± 0.0001	0.0092 ± 0.0002	0.0128 ± 0.0002
Tech5+0104	0.0098 ± 0.0002	0.0138 ± 0.0002	0.0195 ± 0.0002
tid+0008	0.00347 ± 0.00009	0.0056 ± 0.0001	0.0077 ± 0.0001
TigerIT+0066	0.0093 ± 0.0002	0.0158 ± 0.0002	0.0217 ± 0.0002
TigerIT+0067	0.0058 ± 0.0001	0.0131 ± 0.0002	0.0216 ± 0.0002
Pooled	0.0047 ± 0.0001	0.0073 ± 0.0001	0.0109 ± 0.0002

Table 4: *Right index finger FNMRs at various FMRs when cogent+0515 and MINEX III-compliant matchers compare templates created by template generator cogent+0515.*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
006A+0298	0.0054 ± 0.0002	0.0079 ± 0.0002	0.0118 ± 0.0002
0071+0026	0.0035 ± 0.0001	0.0055 ± 0.0002	0.0087 ± 0.0002
aatec+0330	0.0040 ± 0.0001	0.0058 ± 0.0002	0.0087 ± 0.0002
aatec+0450	0.0036 ± 0.0001	0.0053 ± 0.0002	0.0079 ± 0.0002
biowebsolutions+0005	0.0044 ± 0.0001	0.0060 ± 0.0002	0.0079 ± 0.0002
cogent+0515	0.00191 ± 0.00010	0.0024 ± 0.0001	0.0032 ± 0.0001
Decatur+0110	0.0065 ± 0.0002	0.0082 ± 0.0002	0.0111 ± 0.0002
dermalog+0007	0.0048 ± 0.0002	0.0077 ± 0.0002	0.0119 ± 0.0002
dermalog+0010	0.0031 ± 0.0001	0.0052 ± 0.0002	0.0080 ± 0.0002
gemalto+A108	0.0058 ± 0.0002	0.0081 ± 0.0002	0.0114 ± 0.0002
griaule+0112	0.0050 ± 0.0002	0.0074 ± 0.0002	0.0107 ± 0.0002
griaule+0115	0.0037 ± 0.0001	0.0046 ± 0.0002	0.0061 ± 0.0002
id3+14A1	0.0048 ± 0.0002	0.0069 ± 0.0002	0.0100 ± 0.0002
id3+14A2	0.0026 ± 0.0001	0.0035 ± 0.0001	0.0045 ± 0.0002
imagid+0010	0.0041 ± 0.0001	0.0059 ± 0.0002	0.0086 ± 0.0002
imagid+0220	0.0034 ± 0.0001	0.0048 ± 0.0002	0.0071 ± 0.0002
innovatrics+0025	0.0021 ± 0.0001	0.0030 ± 0.0001	0.0046 ± 0.0002
innovatrics+0026	0.0020 ± 0.0001	0.0027 ± 0.0001	0.0039 ± 0.0001
mantrasoftech+0001	0.0042 ± 0.0001	0.0069 ± 0.0002	0.0112 ± 0.0002
miaxis+0028	0.0029 ± 0.0001	0.0044 ± 0.0001	0.0063 ± 0.0002
morpho+0119	0.00165 ± 0.00009	0.00177 ± 0.00009	0.0020 ± 0.0001
morpho+0120	0.00167 ± 0.00009	0.00177 ± 0.00009	0.00196 ± 0.00010
nec+8211	0.0036 ± 0.0001	0.0048 ± 0.0002	0.0064 ± 0.0002
Neurotechnology+0117	0.00173 ± 0.00009	0.0022 ± 0.0001	0.0030 ± 0.0001
Neurotechnology+020A	0.0046 ± 0.0002	0.0070 ± 0.0002	0.0105 ± 0.0002
omnigarde+0003	0.0036 ± 0.0001	0.0058 ± 0.0002	0.0090 ± 0.0002
omnigarde+0010	0.0031 ± 0.0001	0.0050 ± 0.0002	0.0076 ± 0.0002
Pentacomp+0001	0.0047 ± 0.0002	0.0077 ± 0.0002	0.0119 ± 0.0002
roc+0009	0.0046 ± 0.0002	0.0065 ± 0.0002	0.0099 ± 0.0002
ssb+0040	0.0051 ± 0.0002	0.0078 ± 0.0002	0.0115 ± 0.0002
startek+0041	0.0051 ± 0.0002	0.0072 ± 0.0002	0.0099 ± 0.0002
Tech5+0104	0.0066 ± 0.0002	0.0093 ± 0.0002	0.0135 ± 0.0003
tid+0008	0.0029 ± 0.0001	0.0045 ± 0.0002	0.0059 ± 0.0002
TigerIT+0066	0.0069 ± 0.0002	0.0115 ± 0.0002	0.0161 ± 0.0003
TigerIT+0067	0.0049 ± 0.0002	0.0105 ± 0.0002	0.0179 ± 0.0003
Pooled	0.0038 ± 0.0001	0.0055 ± 0.0002	0.0081 ± 0.0002

Table 5: *Left index finger FNMRs at various FMRs when cogent+0515 and MINEX III-compliant matchers compare templates created by template generator cogent+0515.*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
006A+0298	0.0087 ± 0.0002	0.0135 ± 0.0003	0.0203 ± 0.0003
0071+0026	0.0053 ± 0.0002	0.0093 ± 0.0002	0.0148 ± 0.0003
aatec+0330	0.0059 ± 0.0002	0.0093 ± 0.0002	0.0141 ± 0.0003
aatec+0450	0.0050 ± 0.0002	0.0081 ± 0.0002	0.0132 ± 0.0003
biowebsolutions+0005	0.0059 ± 0.0002	0.0089 ± 0.0002	0.0125 ± 0.0003
cogent+0515	0.00192 ± 0.00010	0.0029 ± 0.0001	0.0044 ± 0.0001
Decatur+0110	0.0107 ± 0.0002	0.0137 ± 0.0003	0.0186 ± 0.0003
dermalog+0007	0.0081 ± 0.0002	0.0135 ± 0.0003	0.0208 ± 0.0003
dermalog+0010	0.0055 ± 0.0002	0.0096 ± 0.0002	0.0144 ± 0.0003
gemalto+A108	0.0093 ± 0.0002	0.0140 ± 0.0003	0.0208 ± 0.0003
griaule+0112	0.0083 ± 0.0002	0.0130 ± 0.0003	0.0187 ± 0.0003
griaule+0115	0.0054 ± 0.0002	0.0073 ± 0.0002	0.0106 ± 0.0002
id3+14A1	0.0066 ± 0.0002	0.0101 ± 0.0002	0.0155 ± 0.0003
id3+14A2	0.0030 ± 0.0001	0.0046 ± 0.0002	0.0069 ± 0.0002
imagid+0010	0.0059 ± 0.0002	0.0090 ± 0.0002	0.0138 ± 0.0003
imagid+0220	0.0046 ± 0.0002	0.0074 ± 0.0002	0.0120 ± 0.0002
innovatrics+0025	0.0029 ± 0.0001	0.0051 ± 0.0002	0.0084 ± 0.0002
innovatrics+0026	0.0026 ± 0.0001	0.0043 ± 0.0001	0.0068 ± 0.0002
mantrasoftech+0001	0.0074 ± 0.0002	0.0131 ± 0.0003	0.0225 ± 0.0003
miaxis+0028	0.0039 ± 0.0001	0.0068 ± 0.0002	0.0108 ± 0.0002
morpho+0119	0.00138 ± 0.00008	0.00176 ± 0.00009	0.0023 ± 0.0001
morpho+0120	0.00142 ± 0.00008	0.00177 ± 0.00009	0.0023 ± 0.0001
nec+8211	0.0050 ± 0.0002	0.0077 ± 0.0002	0.0106 ± 0.0002
Neurotechnology+0117	0.00156 ± 0.00009	0.0025 ± 0.0001	0.0039 ± 0.0001
Neurotechnology+020A	0.0072 ± 0.0002	0.0115 ± 0.0002	0.0176 ± 0.0003
omnigarde+0003	0.0056 ± 0.0002	0.0093 ± 0.0002	0.0144 ± 0.0003
omnigarde+0010	0.0050 ± 0.0002	0.0083 ± 0.0002	0.0128 ± 0.0003
Pentacomp+0001	0.0080 ± 0.0002	0.0135 ± 0.0003	0.0215 ± 0.0003
roc+0009	0.0068 ± 0.0002	0.0110 ± 0.0002	0.0180 ± 0.0003
ssb+0040	0.0081 ± 0.0002	0.0134 ± 0.0003	0.0196 ± 0.0003
startek+0041	0.0076 ± 0.0002	0.0112 ± 0.0002	0.0160 ± 0.0003
Tech5+0104	0.0127 ± 0.0003	0.0180 ± 0.0003	0.0254 ± 0.0004
tid+0008	0.0041 ± 0.0001	0.0068 ± 0.0002	0.0098 ± 0.0002
TigerIT+0066	0.0120 ± 0.0002	0.0203 ± 0.0003	0.0271 ± 0.0004
TigerIT+0067	0.0066 ± 0.0002	0.0155 ± 0.0003	0.0252 ± 0.0004
Pooled	0.0056 ± 0.0002	0.0090 ± 0.0002	0.0137 ± 0.0003

Table 6: *Two finger FNMRs at various FMRs when cogent+0515 and MINEX III-compliant matchers compare templates created by template generator cogent+0515.*

Matcher	FNMR @ FMR=0.01	FNMR @ FMR=0.001	FNMR @ FMR=0.0001
006A+0298	0.00060 ± 0.00006	0.00107 ± 0.00007	0.00174 ± 0.00009
0071+0026	0.00012 ± 0.00003	0.00032 ± 0.00004	0.00064 ± 0.00006
aatec+0330	0.00025 ± 0.00004	0.00054 ± 0.00005	0.00090 ± 0.00007
aatec+0450	0.00016 ± 0.00003	0.00036 ± 0.00004	0.00071 ± 0.00006
biowebsolutions+0005	0.00031 ± 0.00004	0.00054 ± 0.00005	0.00083 ± 0.00007
cogent+0515	0.000002 ± 0.000004	0.000009 ± 0.000007	0.00005 ± 0.00002
Decatur+0110	0.00076 ± 0.00006	0.00105 ± 0.00007	0.00154 ± 0.00009
dermalog+0007	0.00018 ± 0.00003	0.00043 ± 0.00005	0.00087 ± 0.00007
dermalog+0010	0.00009 ± 0.00002	0.00033 ± 0.00004	0.00072 ± 0.00006
gemalto+A108	0.00064 ± 0.00006	0.00105 ± 0.00007	0.00165 ± 0.00009
griaule+0112	0.00049 ± 0.00005	0.00083 ± 0.00006	0.00137 ± 0.00008
griaule+0115	0.00022 ± 0.00003	0.00038 ± 0.00004	0.00064 ± 0.00006
id3+14A1	0.00041 ± 0.00005	0.00068 ± 0.00006	0.00101 ± 0.00007
id3+14A2	0.00003 ± 0.00001	0.00013 ± 0.00003	0.00024 ± 0.00003
imagid+0010	0.00023 ± 0.00003	0.00049 ± 0.00005	0.00089 ± 0.00007
imagid+0220	0.00013 ± 0.00003	0.00031 ± 0.00004	0.00065 ± 0.00006
innovatrics+0025	0.000008 ± 0.000007	0.00005 ± 0.00002	0.00015 ± 0.00003
innovatrics+0026	0.000006 ± 0.000006	0.00004 ± 0.00001	0.00014 ± 0.00003
mantrasoftech+0001	0.00029 ± 0.00004	0.00082 ± 0.00006	0.0023 ± 0.0001
miaxis+0028	0.00005 ± 0.00002	0.00019 ± 0.00003	0.00036 ± 0.00004
morpho+0119	0.000002 ± 0.000004	0.000004 ± 0.000005	0.000013 ± 0.000009
morpho+0120	0.000000 ± 0.000003	0.000006 ± 0.000006	0.000009 ± 0.000007
nec+8211	0.00024 ± 0.00004	0.00046 ± 0.00005	0.00070 ± 0.00006
Neurotechnology+0117	0.000000 ± 0.000003	0.000006 ± 0.000006	0.00005 ± 0.00002
Neurotechnology+020A	0.00041 ± 0.00005	0.00075 ± 0.00006	0.00120 ± 0.00008
omnigarde+0003	0.00008 ± 0.00002	0.00025 ± 0.00004	0.00054 ± 0.00005
omnigarde+0010	0.00004 ± 0.00001	0.00017 ± 0.00003	0.00037 ± 0.00004
Pentacomp+0001	0.00054 ± 0.00005	0.00108 ± 0.00007	0.00196 ± 0.00010
roc+0009	0.00048 ± 0.00005	0.00083 ± 0.00006	0.00139 ± 0.00008
ssb+0040	0.00034 ± 0.00004	0.00076 ± 0.00006	0.00134 ± 0.00008
startek+0041	0.00058 ± 0.00005	0.00096 ± 0.00007	0.00145 ± 0.00009
Tech5+0104	0.00058 ± 0.00005	0.00098 ± 0.00007	0.00157 ± 0.00009
tid+0008	0.00012 ± 0.00002	0.00031 ± 0.00004	0.00059 ± 0.00005
TigerIT+0066	0.00063 ± 0.00006	0.00150 ± 0.00009	0.0025 ± 0.0001
TigerIT+0067	0.00029 ± 0.00004	0.00099 ± 0.00007	0.0024 ± 0.0001
Pooled	0.00024 ± 0.00004	0.00049 ± 0.00005	0.00092 ± 0.00007

5 References

[1] Jonathan N. Bradley, Christopher M. Brislawn, and Thomas Hopper. FBI wavelet/scalar quantization standard for gray-scale fingerprint image compression. In *SPIE, Visual Information Processing II*, 1961. 4

[2] George Doddington, Walter Liggett, Alvin Martin, Mark Przybocki, and Douglas Reynolds. Sheep, goats, lambs and wolves a statistical analysis of speaker performance in the nist 1998 speaker recognition evaluation. In *INTERNATIONAL CONFERENCE ON SPOKEN LANGUAGE PROCESSING*, 1998. 4

[3] Patrick Grother Elham Tabassi, George W. Quinn. When to fuse two biometrics. In *IEEE Computer Society on Computer Vision and Pattern Recognition, Workshop on Multi-Biometrics*, 2006. 4

[4] Robert Fontana, Giovanni Pistone, and Maria Rogantin. Classification of two-level factorial fractions. *Journal of Statistical Planning and Inference*, 87:149–172, 2000. 4

[5] P. Grother, M. McCabe, C. Watson, M. Indovina, W. Salamon, P. Flanagan, E. Tabassi, E. Newton, and C. Wilson. Performance and Interoperability of the INCITS 378 Fingerprint Template. Technical report, NIST, 2006. 5

[6] Olaf Henniger and Dirk Scheuermann. Minutiae template conformance and interoperability issues. In Arslan Brömme, Christoph Busch, and Detlef Hühnlein, editors, *BIOSIG*, volume 108 of *LNI*, pages 25–32. GI, 2007. 12

[7] A. Martin, G. Doddington, T. Kamm, M. Ordowski, and M. Przybocki. The DET curve in assessment of detection task performance. In *Proc. Eurospeech*, pages 1895–1898, 1997. 4

[8] George W. Quinn. Evaluation of latent fingerprint technologies: Fusion. In *NIST Latent Fingerprint Testing Workshop Recognition, Workshop*, 2009. 4

[9] Elham Tabassi, Patrick Grother, Wayne Salamon, and Craig Watson. Minutiae interoperability. In Arslan Brömme, Christoph Busch, and Detlef Hühnlein, editors, *BIOSIG*, volume 155 of *LNI*, pages 13–30. GI, 2009. 12, 14

[10] Edwin B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. 4